

Samarth Galchar

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EDUCATION

Khoury College of Computer Sciences, Northeastern University - Boston, USA Jan 2024 - May 2026
Master of Science in Computer Science | GPA: 3.7/4.0
Courses: Algorithms, Programming Design Paradigm, Database Systems, Machine Learning, MLOps, Computer Vision
Charotar University of Science and Technology - Gujarat, India Aug 2019 - Jun 2022
Bachelor of Technology in Computer Engineering

TECHNICAL SKILLS

Core CS: Data Structures & Algorithms, Object-Oriented Design, Design Patterns, Complexity Analysis
Languages: Python, Java, TypeScript, JavaScript, C++, C, SQL
Backend & Web: REST APIs, Unit testing, Microservices, Node.js, Express.js, React.js, Angular, HTML5/CSS3
Data & Cloud: MySQL, PostgreSQL, MongoDB, Redis, AWS, GCP, Docker, Git, CI/CD (GitHub Actions)
AI & ML: PyTorch, Scikit-learn, Tensorflow, LLM/RAG systems, LangChain, LangGraph, HuggingFace,

WORK EXPERIENCE

Cognitus - Dallas, TX Jul 2025 - Jan 2026
Full Stack Developer Intern | *Vue.js, Node.js, PouchDB, REST*

- Delivered features and debugged production issues across modules of an internal ROM estimation tool, iterating on async Vue.js UI/UX improvements driven by stakeholder feedback
- Built and extended Node.js REST endpoints to support new estimation workflows and data persistence
- Integrated PouchDB for offline-first data storage and synchronization, improving fault tolerance for internal users
- Completed an internal **Generative AI certification** covering LLM-assisted development workflows

Simform Solutions - Gujarat, India Jun 2022 - Aug 2023
Full stack Developer | *React.js, Angular, Node.js, MongoDB*

- Developed scalable microservices with Node.js and MongoDB to accelerate REST API call processing
- Optimized database interactions to improve query efficiency and preserve data integrity under load
- Contributed to code reviews, version control, and agile CI/CD workflows on a cross-functional team

Charotar University of Science and Technology - Gujarat, India Apr 2020 - Sep 2020
Machine Learning Research Intern | *Python, NumPy, Jupyter Notebook, NLTK, Gensim*

- Analyzed and applied word embedding techniques (Word2Vec) to evaluate syntactic and semantic similarity preservation
- Enhanced syntactic (0.46) and semantic (0.38) similarity scores over Skip-gram and FastText baselines
- Documented experiments and synthesized insights to support academic publications in Natural Language Processing

PROJECTS

Remaining Useful Life Estimation for Turbofan Engines | *Python, PyTorch, Scikit-learn* Feb - Apr 2026

- Conducted a comparative study of Ridge Regression, XGBoost, 1D CNN, and LSTM on NASA C-MAPSS (FD001, FD003), evaluating both raw sensor sequences and hand-crafted engineered features
- Trained an LSTM on raw sequences predicting remaining engine life within ~14-15 cycles of failure, outperforming prior benchmarks; XGBoost matched this within ~13 cycles using engineered features
- Demonstrated temporal dependency through hidden state visualization and sequence length sensitivity analysis

AI-thelete | *FastAPI, LangChain, Google Cloud, Docker, Streamlit* Jan - Apr 2025

- Architected a Retrieval-Augmented Generation chatbot with FastAPI, LangChain, and the OpenAI API, persisting agent state via LangGraph checkpointing and versioned prompts across sessions
- Enabled context-aware retrieval with Pinecone and HuggingFace embeddings, grounding responses in real user data
- Containerized services with Docker and deployed on GCP with automated redeployments via GitHub Actions

Kanbas – Learning Management System | *MERN Stack, AWS S3, REST* May - Aug 2024

- Shipped a scalable LMS on the MERN stack with RESTful APIs, role-based authentication, and AWS S3 storage
- Implemented quiz functionality with conditional access control enforced server-side per user role

Portfolio Simulator | *Core Java, JUnit, Swing, MVC* Mar - Apr 2024

- Built a Java desktop application managing 100+ stock portfolios, applying MVC and object-oriented design patterns
- Integrated the Alpha Vantage API for real-time quotes, handling call limits to sustain retrieval across 25 tickers
- Computed portfolio performance metrics and packaged the app as an executable JAR for cross-platform deployment